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DEPARTMENT OF DEFENSE HANDBOOK

DESIGN FOR TESTABILITY (DFT) FOR BOUNDARY SCAN DIAGNOSTICS (BSD)

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FOREWORD

1. This handbook is approved for use by the Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense (DoD).

2. This handbook provides guidance on the acquisition of Boundary Scan, or JTAG, technology as part of a larger acquisition of electronic equipment.

3. This handbook is intended to aid DoD personnel in writing requirements for Boundary Scan, or JTAG, technologies as part of a system performance specification.

4. Comments, suggestions, questions or additional information on this document should be addressed to: Naval Air Warfare Center Aircraft Division, Code 4.1.2.2, Route 547, Mail Stop 120-3, Joint Base MDL, NJ 08733-5100 or by email to michael.sikora@navy.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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1. SCOPE

1.1 Scope. This handbook provides guidance for the Department of Defense (DoD) to acquire avionics systems that use Boundary Scan. This document describes the design considerations necessary for a host system to implement Boundary Scan Diagnostics (BSD). Boundary Scan is a technique used to test the interconnections between integrated circuits (ICs) on a printed circuit boards (PCBs), as well as to test the health of the ICs themselves and to load data into them. Boundary Scan can also be used in a system-level configuration between several boards. Boundary Scan is low cost and requires less physical access to circuit components than traditional testing techniques, such as In-Circuit Testing (ICT) for detecting and isolating component failures. With today's highly complex and dense circuit cards, Boundary Scan is a useful technique for troubleshooting and ensuring production quality.

The Joint Test Action Group (JTAG), an electronics industry association formed in 1985, has used Boundary Scan to develop the Institute of Electrical and Electronic Engineers (IEEE) Standard 1149.1-1990, titled "Standard Test Access Port and Boundary-Scan Architecture." This IEEE standard has become synonymous and interchangeable with the group's name. Therefore, this handbook will use the terms "Boundary Scan" and "JTAG" interchangeably. Boundary Scan/JTAG can be used at various levels of maintenance for testing avionic circuit cards. DoD can benefit from this technology as it will enhance the reliability, availability, and maintainability (RAM) of avionics and, as a result, increase operational availability (Ao) and potentially reduce the total ownership cost (TOC) of systems acquired.

Boundary Scan can test the interconnections of ICs such as Field Programmable Gate Arrays (FPGAs) and Central Processing Units (CPUs) on a PCB for short and open circuit faults. Data is sent to the IC from a piece of Automatic Test Equipment (ATE), or from a Personal Computer (PC). The output of the IC is sent back to the ATE to be examined. With multiple ICs on a PCB, the ATE sends the boundary scan test signal into the first IC and the output of this IC will be the input of the next IC, the output of the second IC will be the input to the third IC, and so on. The output of the last IC on the PCB will be sent back to the ATE for examination, thus creating a chain, also known as the "scan chain." A two-IC configuration example is shown on Figure 1. The scan chain verifies that the Boundary-Scan-enabled ICs on the PCB are communicating properly and identifies any short-circuit or open-circuit failures on the PCB.

Boundary Scan can also be used in a system-level test configuration, connecting multiple boards. Just as boundary scan can connect multiple ICs to form a scan chain, system-level boundary scan can form a chain between multiple boards, allowing testing interconnections across multiple boards using only one source. System-level boundary scan can be configured in many different topologies. Some of the proposed topologies include a ring configuration, where the scan chains of each board are each chained; this higher level scan chain would be controlled by a single boundary scan controller. Another proposed topology is a star configuration, the controller would have access to the scan chains of all the boards, but be able to access them in parallel instead of in series. One more proposed topology has a system-level controller's boundary scan connections across the backplane connecting all the boards together. Each board's boundary scan controller interfaces with this bus, and would then provide boundary scan testing to individual boards. No matter the configuration, system-level boundary scan provides benefits in system

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testability, as testing would be standardized across an entire system, and only a single test sequence would be needed.

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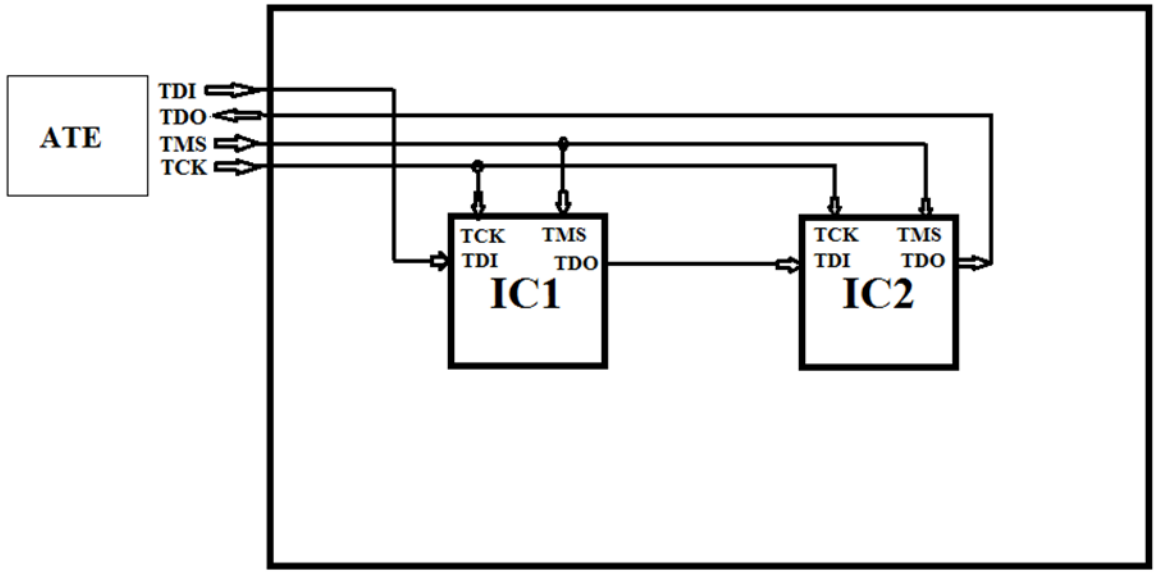


FIGURE 1. Boundary scan chain.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed below are not necessarily all of the documents referenced herein, but are those needed to understand the information provided by this handbook.

2.2 Government documents.

2.2.1 Handbook. The following handbook forms a part of this document to the extent specified herein.

DEPARTMENT OF DEFENSE

HANDBOOK

MIL-HDBK-2165 - Testability Program for Systems and Equipments

(Copies of this document are available at http://quicksearch.dla.mil.)

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2.3 Non-Government publications. The following non-government documents form a part of this document to the extent specified herein.

INSTITUTE OF ELECTRICAL AND ELECTRONIC ENGINEERS (IEEE)

IEEE 1149.1	-	Standard Test Access Port and Boundary-Scan Architecture
IEEE 1532	-	Standard for Boundary-Scan-based In System Configuration of Programmable Devices
IEEE 1687	-	Standard for Access and Control of Instrumentation Embedded within a Semiconductor Device

(Copies of these documents are available from http://ieee.org.)

ELECTRONIC INDUSTRIES ALLIANCE (EIA)

IEC 61690-2	-	Electronic Design Interface Format (EDIF) Version 4.0.0

(Copies of this document are available from http://www.iec.ch/.)

3. ACRONYMS AND DEFINITIONS

The following acronyms are applicable to this handbook.

A_o	Operational Availability
ASIC	Application-Specific Integrated Circuit
BOM	Bill of Materials
BR	Bypass Register
BSD	Boundary Scan Diagnostics
BSR	Boundary Scan Register
CPU	Central Processing Unit
DFT	Design for Test
DoD	Department of Defense
EDIF	Electronic Design Interchange Format
FPGA	Field Programmable Gate Array
FSM	Finite State Machine
I-Level	Intermediate Level of Maintenance
IC	Integrated Circuit
IJTAG	Internal Joint Test Action Group
I/O	Input/Output
IR	Instruction Register
JTAG	Joint Test Group
O-Level	Organizational Level of Maintenance
PCB	Printed Circuit Board

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SRA	Shop Replaceable Assembly
TAP	Test Access Port
TCK	Test Clock
TDI	Test Data Input
TDO	Test Data Output
TDR	Test Data Register
TMS	Test Mode Select
TRST	Test Reset
WRA	Weapons Replaceable Array

4. GENERAL GUIDANCE

4.1 Boundary scan overview. Boundary-scan-based DFT architectures have become inherent in modern electronics design. Boundary scan consists of four (five in some cases) signals that comprise a Test Access Port (TAP). The four signals are: Test Data Input (TDI), which is the input data inserted from the ATE into ICs on a PCB; Test Data Output (TDO), which is the output data from the ICs that gets sent back to the ATE for examination; Test Clock (TCK), which synchronizes the transfer of data and instructions among various registers on the ICs; and Test Mode Select (TMS), which acts as the sole test control input to the TAP controller. The optional fifth signal is the Test Reset (TRST), which resets the TAP Controller. An example of this Boundary Scan TAP architecture is shown on Figure 2.

Boundary Scan adds linked cells at the boundary of the IC’s at each input and output pins, thus creating a register known as the Boundary Scan Register (BSR). These are the pins that the TAP signals are inserted into which test the interconnections of the ICs.

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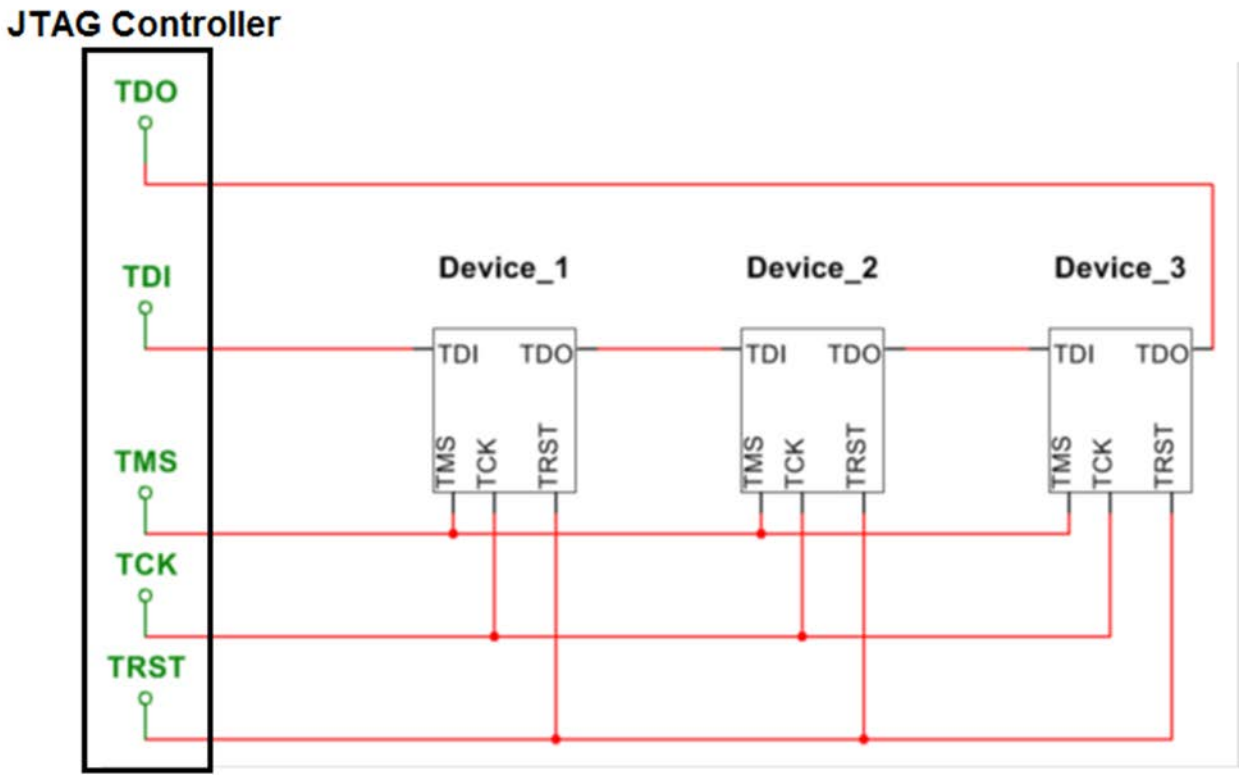


FIGURE 2. Boundary scan TAP architecture example.

5. DETAILED GUIDANCE

5.1 System design. A single TAP, ideally, can be used for both boundary scan diagnostics and programming. Therefore, standard industry practice is to avoid a design that has multiple TAPs on a single board, as it will unduly complicate the boundary scan chain. However, multiple TAPs can be used for a board or system that requires a separation of testing and programming ports for security purposes.

5.1.1 Printed circuit boards (PCB) scan chain. Organize all PCBs in a system supporting boundary scan into a single scan chain. It is preferable that there is a single TAP for a single boundary-scan device, and that it will operate through this single TAP.

5.1.1.1 Multiple scan chains. Use multiple TAPs for systems whose complexity exceeds the ability to conduct boundary scan on a single scan chain. All the TAPs should be accessible by the BSD device.

5.1.1.1.1 One TAP per PCB. For systems that include multiple circuit boards, there should be only one TAP on each boundary-scan-enabled PCB.

5.1.2 TAP connection(s) accessibility. All boundary scan TAP connections should be available on an external connector of a Weapons Replaceable Assembly (WRA) for Operational

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Level (O-level) of maintenance testing, and/or accessible on a Shop Replaceable Assembly (SRA) at the Intermediate Level (I-level) of maintenance on an edge connector. The connector should be appropriate to the intended operational environment.

5.2 Configuration Management (CM). A product, throughout its lifecycle, may be updated or redesigned several times. Boundary-scan-enabled boards each have a unique infrastructure and interconnections that are dependent on the specific internal components. Thus, each revision should be maintained and tracked by a BSD test plan separately in a configuration management. Track each configuration change affecting the BSD scan chain in configuration management, as applicable.

5.2.1 Data from PCB manufacturer. Ensure that each PCB is delivered with its netlist, schematic, Bill of Materials (BOM) and associated Boundary Scan Description Language (BSDL) files stored under configuration management.

5.2.2 Tracking of PCB versions. Configuration management should track each revision of a PCB, including its netlist, schematic, BOM and associated BSDL files.

5.2.3 BSD data formatting. Format each netlist, schematic, BOM, and associated BSDL files in an industry standard format for that type of file. One such industry standard format is the Electronic Design Interchange Format (EDIF) for netlists and schematics.

5.3 Routing. This section describes the boundary scan routing for the cases where boards must support device programming (as outlined in IEEE 1532), board emulation or fault detection.

5.3.1 BSD devices signal routing. All IEEE 1149.1 signals should be routed between all boundary scan enabled devices.

5.3.2 BSD TAP connector routing. All IEEE 1149.1 boundary scan signals should be routed to a TAP connector.

5.3.3 Multi-scan chain routing. Route the boundary scan signals between each TAP connector when multiple scan chains are used. Some emulation solutions require a single device on the chain. Multiple TAPs will allow easier isolation of an infrastructure failure in that case.

5.3.4 Clock skew verification. Verify the signal integrity of a scan chain to ensure the clock skew is in compliance with IEEE 1149.1.

5.3.5 Devices using JTAG port for emulation. Place devices that use the JTAG port for emulation on a separate chain. Separating the emulation scan chain and the non-emulation scan chain precludes emulation issues.

5.4 Boundary Scan Enabled Components. Boundary scan enabled components will need to follow the listed design considerations for a successful BSD implementation.

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5.4.1 Component compliance. The boundary scan component should be compliant with IEEE 1149.1.

5.4.2 Component BSD required BSDL file. Verify that the boundary scan component BSDL file matches the datasheet of the component.

5.4.3 Boundary-scan clock signals. The boundary scan TCK clock signals should follow the same layout convention as other PCB clock signals.

5.5 TAP placement. Boundary scan enabled circuit cards and WRA's will need to follow TAP placement requirements, in the following subsections, to ensure usability.

5.5.1 TAP signal pins separation. The TAP signal pins (TDI, TDO, TMS, TCK and TRST) should not be electrically short-circuited together.

5.5.2 TAP signal pins placement to power. Place TAP signal pins adjacent to power pins on the component, when possible. With fine pitch devices, the most common manufacturing faults are short-circuited adjacent pins or open pins. For example, a TAP signal short-circuited to a power pin is much more easily diagnosed than two TAP signals short-circuited together, or a TAP signal short-circuited to another signal.

5.5.3 TAP signal pins placement to chip corner. Wherever practical, the TAP signal pins should be placed near, or on, the corners of a chip in order to reduce the chance that the pins will be short-circuited to adjacent pins.

5.5.4 Single scan chain. Wherever practical, route all boundary-scan component connections into a single scan chain, as shown on Figure 1.

5.5.4.1 Multiple TAP use. When all of the boundary scan components in a system cannot be organized into a single scan chain, then use more than one TAP.

5.5.5 TAP signals to input/output (I/O) pins. TAP signals should not be connected to I/O pins of devices in the scan chain. JTAG's flexible mode allows TAP signals to be optionally used as I/O pins on some devices. This is to preclude board designers from using TAP signals as extra I/O outside of their intended boundary scan function.

5.6 Higher-order instruction capture bits. Set the higher-order instruction capture bits to a known value so that the component can be easily identified.

5.7 Tri-state and bi-directional pin control. The bidirectional and tri-state pins for direction/high impedance state control should be separated.

5.8 Unused instruction Opcodes. Unused TAP instruction opcodes should default to BYPASS.

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5.9 Connectors, signals, and termination. The design should follow a connection scheme that will enable maintainability of the device.

5.9.1 Board connector. There are no standard JTAG connectors, but there are 10-, 16- and 20-pin configurations. Use a 10-pin connector for Boundary Scan when no direct writing to flash memory is required. This is also the most secure connector, since it does not allow interfaces to access program memory.

5.9.2 TAP connector TDO 33 Ohm resistor. The 33 Ohm series resistor on the last device in the chain should be connected TDO pin. The resistor should be physically located as near as practical to the TDO pin.

5.9.3 Pull-up resistors. Place a 1KOhm pull-up resistor next to any supply voltage (V_{CC}) between 1.25V and 3.3V, matching the boundary-scan devices in the chain.

5.9.4 Grounding. All grounds should be connected and accessible at the TAP.

5.10 Bypassing Boundary Scan Devices. When necessary to bypass a boundary scan device, include additional logic to continue the boundary scan chain.

5.11 Configurable BSDs. A configurable boundary scan device should set boundary scan mode using the TRST or a combination of TCK and TMS signal levels. However, some boundary-scan-enabled devices require additional signals to be set to a defined voltage level in order to be put into boundary scan mode. Other boundary-scan-enabled devices (such as FPGAs) can be configured during power-up and, as a result, behave differently in boundary scan mode. This is what is meant by "configurable boundary scan."

5.12 IJTAG (compatibility of board test instrumentation). Board manufacturers do not use their own intellectual property on a chip such as an FPGA or application-specific integrated circuit (ASIC), but also on those acquired from other suppliers. Thus, internal JTAG (or, IJTAG) specifies the IEEE 1687 standard for interface of embedded instruments (on-chip programmable logic) to the JTAG TAP. The purpose of the IEEE standard is to connect the on-chip instruments and describe a language for communicating with the instruments via IEEE 1149.1 Test Data Registers.

5.13 IEEE 1687 compliance for TAP connection. On-chip instrumentation should conform to IEEE 1687 in order to interface via the JTAG TAP connector.

5.14 Verification. It is recommended that a verification plan be requested by the Government in order to verify boundary scan functions, as per applicable contract requirements.

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6. NOTES

6.1 Intended use. This handbook provides guidance on the acquisition of Boundary Scan, or JTAG, technology as part of a larger acquisition of electronic equipment.

6.2 Subject term (key word) listing.

Avionics
BIT
Built-in Test
I-Level
Joint Test Action Group
JTAG
Maintenance
O-Level
R&M
Reliability & Maintainability
Supportability
Systems Engineering

6.3 Document reference. All references to IEEE 1149.1 refer to the IEEE 1149.1-2013 version.

Custodian:
Navy – AS

Preparing activity:
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